

1 Minimum Functional Dependency Set

#	Functional Dependency
1	Race_ID → Grand_Prix_Name, Circuit_Name, Country, Race_Date
2	Result_ID → Race_ID, Driver_ID, Rank, Time_Taken, Laps, Points
3	Ticket_ID → Race_ID, Seat_Category, Price, Ticket_Availability, Booking_Date, Quantity
4	Driver_ID → Driver_Name, Nationality, Date_of_Birth, Career_Wins, Championships_Won, Points
5	Team_ID → Team_Name, Team_Principal, Base_Location, Championships_Won
6	User_ID → Name, Email, Phone_No, Country, Address
7	Order_ID → User_ID, Order_Date, Shipping_Address, Payment_Service_Name, Payment_Reference_Number, Order_Status
8	Cart_ID → User_ID, Cart_Status
9	Product_ID → Seller_ID, Product_Name, Description, Price, Available_Quantity, Product_Category, Team_ID
10	Seller_ID → Organization_Name, Organization_Address, Contact_Person_Name, Contact_Number, Email
11	Review_ID → User_ID, Product_ID, Rating, Comment, Review_Date
12	League_ID → Created_By, League_Name, Max_Participants, Start_Date, End_Date
13	Fantasy_Team_ID → League_ID, User_ID, Team_Name, Total_Points
14	(Cart_ID, Product_ID) → Quantity
15	(Order_ID, Product_ID) → Quantity, Price_at_Purchase

2 BCNF Proof for All Relations

Every relation in the schema satisfies Boyce-Codd Normal Form (BCNF). For each relation, the left-hand side of every non-trivial functional dependency is a superkey.

#	Relation	Functional Dependency	Primary Key	BCNF Verdict
1	User	<i>User_ID → Name, Email, Phone_No, Country, Address</i>	User_ID	Primary key; is a superkey → BCNF
2	Seller	<i>Seller_ID → Organization_Name, Organization_Address, Contact_Person_Name, Contact_Number, Email</i>	Seller_ID	Primary key; is a superkey → BCNF
3	Team	<i>Team_ID → Team_Name, Team_Principal, Base_Location, Championships_Won</i>	Team_ID	Primary key; is a superkey → BCNF
4	Driver	<i>Driver_ID → Driver_Name, Nationality, Date_of_Birth,</i>	Driver_ID	Primary key; is a superkey → BCNF

#	Relation	Functional Dependency	Primary Key	BCNF Verdict
		<i>Career_Wins, Championships_Won, Points</i>		
5	Race_Event	<i>Race_ID → Grand_Prix_Name, Circuit_Name, Country, Race_Date</i>	Race_ID	Primary key; is a superkey → BCNF
6	Race_Result	<i>Result_ID → Race_ID, Driver_ID, Rank, Time_Taken, Laps, Points</i>	Result_ID / (Race_ID, Driver_ID)	Both are superkeys → BCNF
7	Ticket	<i>Ticket_ID → Race_ID, Seat_Category, Price, Ticket_Availability, Booking_Date, Quantity</i>	Ticket_ID	Primary key; is a superkey → BCNF
8	Merchandise	<i>Product_ID → Seller_ID, Product_Name, Description, Price, Available_Quantity, Product_Category, Team_ID</i>	Product_ID	Primary key; is a superkey → BCNF
9	Orders	<i>Order_ID → User_ID, Order_Date, Shipping_Address, Payment_Service_Name, Payment_Reference_Number, Order_Status</i>	Order_ID	Primary key; is a superkey → BCNF
10	Review	<i>Review_ID → User_ID, Product_ID, Rating, Comment, Review_Date</i>	Review_ID	Primary key; is a superkey → BCNF
11	Fantasy_League	<i>League_ID → Created_By, League_Name, Max_Participants, Start_Date, End_Date</i>	League_ID	Primary key; is a superkey → BCNF
12	Fantasy_Team	<i>Fantasy_Team_ID → League_ID, User_ID, Team_Name, Total_Points</i>	Fantasy_Team_ID	Primary key; is a superkey → BCNF
13	Cart	<i>Cart_ID → User_ID, Cart_Status</i>	Cart_ID	Primary key; is a superkey → BCNF
14	Cart_Item	<i>(Cart_ID, Product_ID) → Quantity</i>	(Cart_ID, Product_ID)	Composite PK; is a superkey → BCNF
15	Order_Item	<i>(Order_ID, Product_ID) → Quantity, Price_at_Purchase</i>	(Order_ID, Product_ID)	Composite PK; is a superkey → BCNF
16	Wishlist	<i>Wishlist_ID → Wishlist_Name, User_ID</i>	Wishlist_ID	Primary key; is a superkey → BCNF
17	Wishlist_Item	<i>(Wishlist_ID, Product_ID) → Date_Added, Priority</i>	(Wishlist_ID, Product_ID)	Composite PK; is a superkey → BCNF
18	Fantasy_Team_Driver	<i>(Fantasy_Team_ID, Driver_ID) — no non-key attributes</i>	(Fantasy_Team_ID, Driver_ID)	All attributes form the key → BCNF
19	Fantasy_Team_Team	<i>(Fantasy_Team_ID, Team_ID) — no non-key attributes</i>	(Fantasy_Team_ID, Team_ID)	All attributes form the key → BCNF
20	Driver_Team	<i>(Driver_ID, Team_ID, Season_Year) — no non-key attributes</i>	(Driver_ID, Team_ID, Season_Year)	All attributes form the key → BCNF